

MINI PROJECT 4

Safety and Environmental Hazard Assessment of the Vinyl Chloride Monomer
(VCM) Production Process

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Project Charter

Purpose and Objectives

The purpose of this project was to assess the safety and environmental performance of the vinyl chloride monomer (VCM) production facility. A rigorous analysis of the process, specifically the T-100 units, was conducted in order to identify any hazards associated with materials, operations, equipment conditions, and environmental releases.

Objectives included identifying environmental hazards, assessing material hazards, evaluating operational hazards, performing a full HAZOP study, recommending engineering controls, and benchmarking the process against AI-generated analysis. Environmental hazards could arise from air emissions, wastewater discharges, and potential solid waste generation. In order to assess material hazards like flammability, toxicity, and incompatibility, the OSHA OSN chemical lists and PEL values were utilized and referenced.

Operational hazards could include overpressure, loss of containment, ignition sources, explosion potential, and equipment degradation. A HAZOP study was conducted to identify deviations in flow, temperature, pressure, and composition in the T-100 units. This information was used to recommend engineering controls and safeguards based on best practices like relief valves, interlocks, alarms, shutdown systems, secondary containment, and flame suppression. Finally, an AI-analysis of the process was tested, and a reflection was formed based off the performance of AI compared to human design.

Roles and Responsibilities

For this project, Project Charter was assigned to Lindsey Meisch, Design Plan was assigned to Khoa Nguyen, Environmental Safety Assessment was assigned to Kyle Hankosky, HAZOP study was assigned to Ahmed Alhassani, and AI and Bibliography were assigned to Kendall Johnson

Timeline

Dates	Milestone
11.26.2025-11.29.2025	Review PFD and Stream Summary Table, assign roles, and outline project deliverables and design plan.
11.29.2025-12.04.2025	Material, environmental, and operational hazard evaluation. Execute HAZOP study for T-100 nodes.
11.29.2025-12.01.2025	Run AI analysis.
12.01.2025-12.04.2025	Compile sources and format final report.
12.05.2025	Submit final report.

Design Plan

Summary of the vinyl-chloride production process

For the process to produce vinyl chloride (VCM), stream 1, containing dry ethylene, and chlorine in stream 2, first enter a direct chlorination reactor (R-100) at 25 °C and 1.5 atm. These two streams then react to form streams 3 and 4, which contain mostly 1,2-dichloroethane (EDC). A condenser (E-100) and reactor pump (P-100) were used to maintain an appropriate liquid level in the reactor and to return the condensed EDC. Stream 5 then delivers the crude EDC to E-101, which uses high-pressure steam to evaporate the EDC. The heated EDC is then sent to the pyrolysis furnace (F-100), which uses natural gas to operate at 500°C and 26 atm to thermally crack the EDC into VCM, hydrochloric acid (HCL), and unreacted EDC. The product stream 7 is then rapidly quenched and separated in V-100, in which stream 8 is the gas stream and stream 9 is the liquid stream. The liquid stream is then cooled in E-102 with cold water before being recycled back into the separator, and the gas stream is partially condensed in E-103 before a valve is opened to send the condensed gas stream into the HCL column T-100. In the distillate, pure HCL is condensed and collected as a by-product in stream 11. The bottoms, containing mainly VCM and EDC in stream 12, are reheated by the reboiler E-105; a portion is recycled as reflux, and the rest is sent to the VCM column T-101. In the distillate section of T-101, high-purity VCM is condensed in E-106 using cold water, which is mainly collected in the reflux drum V-102, and collected as product in stream 14 while a small portion of the product is returned as reflux via P-103. The bottoms section of T-101 contains mostly unreacted EDC, which is cooled and then recycled back to the reactor to improve ethylene conversion.

Outline of specific focus areas for safety and environmental analysis

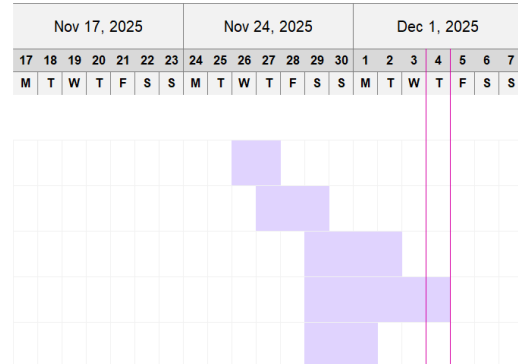
The process that happens around the T-100 unit involves mainly the separation of HCL from the cracked EDC/VCM mixture. The key units of this process are the T-100 distillation column, the overhead condenser E-104, reflux drum V-101, and their associated pumps. The reboiler E-105 as well as all connected piping and valves are also involved with this process. The safety and environmental analysis should be emphasized on equipment that handle acidic and chlorinated streams where corrosion, emissions, and overall material compatibility should be the main concern. The corresponding utilities that the listed units use, such as cooling water, steam, and air, should also be reviewed to ensure safe operations. Environmental analysis for this section should also address potential air releases from overhead systems, water contamination from heat exchanger leaks, and generation of hazardous waste from the units. Similar to T-100, the processes taking place around T-101 involve the separation of EDC and VCM, as well as the purification of VCM. These processes include the T-101 distillation column, overhead condenser E-106, reflux drum V-102, the pump P-103, and the reboiler E-107. Since this section handles high-purity VCM and unreacted EDC, the analysis should focus on the flammability and toxicity of the chemicals present. Utilities such as cooling water, steam, and electricity should also be analyzed to ensure proper functioning of equipment as well as safe working conditions. Environmental safety analysis of this section should focus on potential VCM emissions, leaks from pumps, valves, and heat exchangers, and waste generation during equipment cleaning.

Gantt chart outlining project timelines

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Project start: Wed, 11/26/2025

Task	Assigned To	Progress	Start	End
Initiation				
Project Charter	Lindsey Meisch	100%	11/26/25	11/27/25
Design Plan	Khoa Nguyen	100%	11/27/25	11/29/25
Environmental and Safety Assessment	Kyle Hankosky	100%	11/29/25	12/2/25
HAZOP Study	Ahmed Alhassani	100%	11/29/25	12/4/25
AI Integration & Bibliography	Kendall Johnson	100%	11/29/25	12/1/25



Environmental Safety Assessment

Environmental Issues

Hydrogen chloride, vinyl chloride, and 1,2-dichloroethane are all chemicals being used in T-100.

Reviewing the hydrogen chloride safety data sheet (SDS) (Sigma-Aldrich, Safety Data Sheet, 2024), potential environmental concerns are letting the product enter the drains. This can be mitigated with not having drain access close to units that have hydrogen chloride, not disposing of waste from the unit down the drain, and having a closed loop system. Wastewater generated that potentially contains hydrogen chloride can either be pH checked and neutralized, or go through a variety of membrane or desalination processes.

Reviewing the vinyl chloride SDS_(Sigma-Aldrich, Safety Data Sheet, 2024), potential environmental concerns are letting the product enter the drains. This can be mitigated with not having drain access close to units that have vinyl chloride, not disposing of waste from the unit down the drain, and having a closed loop system. Wastewater generated can be treated according to Patent CN100441528C (China Patent No. CN100441528C, 2005), which involves a multi-stage oxidation adsorption and then carbon filtration process. Note: 1,2-dichloroethane is combustible with carbon oxides according to (Sigma-Aldrich, Safety Data Sheet, 2025), which may pose a problem.

Reviewing the 1,2-dichloroethane SDS (Sigma-Aldrich, Safety Data Sheet, 2025), potential environmental concerns are letting the product enter the drains and the risk of explosion. This can be mitigated by not having drain access close to units that have hydrogen chloride, not disposing of waste from the unit down the drain, and having a closed-loop system.

Additional work will need to be done to ensure thorough separation of hydrogen chloride gas from any waste containing 1,2-dichloroethane as 1,2-dichloroethane combusts with hydrogen chloride. Treat wastewater with

Process Hazards and Safety Compliance

When evaluating the hazard associated with the production of vinyl chloride the materials in the process must be first identified along with their toxicity, flammability instability and regulatory classifications. The following two tables quickly summarize the exposure limits, reportable quantities, and NFPA hazards ratings. These values, particularly the permissible exposure limits are the primary deciding factor in process safety and compliance with state and federal regulations.

Component	Exposure Limit [PPM/8hrs]	Reportable Quantity [lbs]	DOT Shipping Class
Ethylene	200	N/A	2.1 Flammable Gas
Chlorine	0.5	100	2.3 Inhalation Hazard, 5.1 Oxidizer, 8 Corrosive
1,2-dichloroethane	10	100	3 Flammable Liquid, 6.1 Toxic
Vinyl chloride	1	43	2.1 Flammable Gas
Hydrogen Chloride	5	5000	8 Corrosive

NFPA	Health	Flammability	Instability	Physical Hazards
Ethylene	1	4	1	N/A
Chlorine	4	0	0	OX
1,2-dichloroethane	3	3	0	N/A
Vinyl chloride	2	4	3	N/A
Hydrogen Chloride	3	0	0	N/A

The combination of toxic, flammable and highly reactive chemicals in the vinyl chloride production process creates a hazard profile that must be controlled through both engineering and oversight. Chlorine gas, hydrochloric acid and vinyl chloride each have very low PEL's and preset the most severe toxicity risks, requiring fully sealed equipment made of corrosion resistant materials with environmental monitoring to prevent operator exposure. At the same time ethylene, vinyl chloride and 1,2-dochloroethane have high flammability ratings and are in operations at 500C and 26 psi, these operating parameters significantly increase the likelihood of fire and vapor cloud explosions from a loss of containment. These temperature and pressure conditions also amplify reactivity hazards from exothermic chlorination reactions, where failure

of removal of heat can trigger thermal runaway of the unit or over pressurization. Mechanical integrity of the unit is a constant cause for concern because chlorine gas and hydrochloric acid corrode carbon steel meaning that thinning of the wall of the piping in the process could cause a major release.

To manage these risks the process relies on multiple layers of protection to ensure safe operation. Pressure relief devices properly sized per ASME codes and routed to burn towers or “flares” to prevent vessels from rupturing during abnormal process conditions. While automated shutdown systems can intervene when deviations occur. Nitrogen inerting of the process helps mitigate risk of explosion, fixed lower explosion limit “LEL’s” and environmental sensors provide early warnings of leaks, given the low exposure limits of the species present in the process. These engineering measures are supported by operational safeguards required under OSHA PSM, including operator training, well-documented operating procedures and management of change of the unit. Emergency response procedures, self-contained breathing apparatus “SCBA” availability and routine inspections of corrosion-prone areas and equipment ensure that the process can be operated safely while protecting workers and the environment.

HAZOP Study

Parameter	Deviation	Causes	Consequences	Safeguards	Actions
Flow	No Flow	The upstream pump stops operating, and the control valve reaches its maximum closed position or solid particles, or polymer blockages occur, and instrument air supply fails.	It loses separation ability, and the tray becomes dry while producing substandard overhead and hot spots form.	The system includes three main functions which are feed flow control with low-flow alarm and pump status alarm and upstream level control.	The system requires a low-low feed-flow trip to start safe column shutdown operations while pump maintenance uses a bypass line, and predictive maintenance should be applied to pumps and strainers.
Flow	More Flow	The flow control valve remains open while the controller fails, and operators set incorrectly	The system experiences column flooding which results in separation efficiency loss, increased	The system includes two main alarms which trigger when flow rates exceed normal	The system requires a high-high feed-flow interlock to activate valve closure or pump

		high flow rates and start the wrong pump.	internal pressure drop, elevated chances of relief valve activation, and hydrocarbon/ chlorinated vent release to the flare system.	limits and when column pressure differences exceed safe levels and when the relief valve opens to release gas into the flare.	shutdown while access to control systems remains restricted, and operators receive training with written procedures.
Temperature	Too Low	The upstream cooler operates beyond its designed capacity, and the reactor's outlet temperature remains below its setpoint, and the utility conditions do not match the system requirements.	The column fails to vaporize its contents properly, which results in decreased relative volatility and produces off-spec products through degraded separation performance.	The system includes temperature monitoring and alarm functions at the feed point, and it maintains temperature control in the upstream section.	The system requires a low-temperature alarm that will trigger feed valve closure when activated; the feed preheat system design needs evaluation to prevent sub-cooling conditions.
Temperature	Too High	The heater control system fails, and the cooling system loses its function in the upstream section and exothermic reactions become unstable in the reactor.	The feed line experiences premature vaporization and partial flashing which causes column pressure to rise while putting stress on internal equipment and increasing VCM vapor concentrations that raise fire danger levels.	The system operates through temperature control loops and pressure control systems that monitor the column.	The system needs a high-high feed-temperature trip function while implementing insulation and fireproofing measures and verifying the maximum allowable working temperature (MAWT) for

					all construction materials.
Composition	Wrong Composition	Incorrect feed tank selection and oxygen, water, non-condensable contamination, and unreacted chlorine gas escaping from the system.	The system faces three major issues because HCl reacts with water and causes corrosion, VCM and EDC undergo polymerization and phosgene formation becomes possible when CO exists in the system while producing off-spec products and toxic substances.	The system includes feed analyzers with GC technology and upstream interlocks and material identification and labeling systems.	The system requires an online composition analyzer with alarm functions and positive line-up procedures and key-lock systems for manual valve operation and oxygen analyzer installation when air entry becomes possible.

The table above shows the T-100 distillation unit feed section entry point. The process depends on this section because any changes in flow rates or temperatures or pressures, or chemical compositions will cause the entire column to become unstable. The analysis shows that pump failures and control-valve malfunctions and contamination events and temperature and pressure variations create significant threats to the system. The system faces multiple risks because it produces off-spec products and separation efficiency decreases, and dangerous situations occur that could lead to overpressure and toxic substance releases and polymer formation. The study recommends additional safety measures for improved operational safety because the current alarm system and control loops offer basic protection.

AI Integration

The AI model used for this section was ChatGPT 5.1

Prompt: Using the provided process and pictures, could you help me do the following tasks:

- **Environmental Issues**
 - Identify and detail potential environmental concerns, including sources of air emissions, wastewater discharges, and any solid waste production.
 - Propose design features and engineering controls to mitigate the impacts of these discharges. This may include pollution control equipment, wastewater treatment options, or closed-loop systems.
- **Process Hazards and Safety Compliance**
 - Outline and evaluate material and process hazards, adhering to legal and ethical standards. Identify all OSHA Process Safety Management (PSM) highly hazardous materials and include their permissible exposure limits (PELs).
 - Assess risks associated with toxic materials, explosive conditions, and extreme temperatures and pressures, providing recommended safeguards in the design. Examples may include pressure relief systems, automated shutdowns, containment barriers, and emergency response measures.

HAZOP Study

- Conduct a Hazard and Operability (HAZOP) study, focusing on the T-100 units to examine potential deviations in process parameters (e.g., flow, temperature, pressure).
- Apply HAZOP guidewords (e.g., “More,” “Less,” “None”) to assess risks, identify potential causes and consequences, review existing safeguards, and propose further safety recommendations.

Results:

Environmental Issues

- **Potential environmental concerns**
 - Air emissions from T-100 overhead and relief devices containing hydrogen chloride, vinyl chloride, and 1,2-dichloroethane, especially during startups, shutdowns, or relief events.
 - Fugitive emissions from flanges, pump seals, valves, and sampling points around T-100, V-101, E-104, and E-105 carrying toxic and/or carcinogenic chlorinated species.
 - Wastewater contamination from cooling-water leaks in E-104 and E-105 that could introduce HCl and chlorinated organics into the cooling system and, ultimately, into plant effluents.
 - Solid and sludge wastes generated from column cleaning, spent corrosion coupons, degraded internals, and neutralization residues containing chlorinated organics.
- **Design features and engineering controls**
 - Routing all T-100 overhead vents and pressure-relief streams to a dedicated HCl/organics scrubbing system followed by a flare or thermal oxidizer to minimize air emissions.
 - Using double-mechanical-seal pumps, low-emission valves, and an LDAR (leak detection and repair) program for equipment in HCl/VCM/EDC service.

- Providing online monitoring and isolation capability for cooling-water return lines from E-104 and E-105, with diversion to wastewater treatment if organics or low pH are detected.
 - Implementing closed-drain systems and segregated hazardous-waste collection for any liquids drained from T-100, V-101, and associated piping, followed by appropriate neutralization and treatment.
-

Process Hazards and Safety Compliance

- **Material and process hazards**

- T-100 handles OSHA PSM-covered highly hazardous chemicals, including hydrogen chloride, vinyl chloride, and 1,2-dichloroethane, all of which are toxic and corrosive or flammable.
- The column operates at elevated pressure and temperature relative to atmospheric conditions, with hot reboiler circuits and pressurized condensers that can create overpressure, jet-release, and burn hazards.
- Dense, heavier-than-air chlorinated vapors can accumulate in low areas around the T-100 structure, posing asphyxiation, toxic-exposure, and delayed ignition risks.

- **Risk assessment and safeguards**

- Identifying exposure limits and ensuring compliance through engineering controls, ventilation, gas detection, and administrative procedures in the T-100 operating area.
- Providing appropriately sized pressure-relief devices on T-100, V-101, E-104, and E-105, with discharge to a controlled disposal system.
- Implementing automated shutdowns and interlocks tied to high column pressure, abnormal temperatures, low cooling-water flow, and high liquid levels in V-101 and the column base.
- Using suitable materials of construction and corrosion-allowance, periodic inspection, and thickness monitoring for all equipment handling HCl and chlorinated organics.
- Establishing emergency response plans, including local eyewash/showers, spill-containment provisions, and clear evacuation routes around the column structure.

HAZOP Study

- **Scope and nodes**
 - Conduct a Hazard and Operability (HAZOP) study using T-100 as the central system, defining nodes such as: feed to T-100 from V-100/E-103, the T-100 column shell, overhead system (E-104 and V-101), reboiler loop (E-105), and reflux/product pump circuits (P-102, P-104 and connected piping).
- **Parameters and deviations**
 - Examine deviations in key process variables—flow, temperature, pressure, level, and composition—using standard guidewords such as *More*, *Less*, *None*, *Reverse*, and *As well as*.
 - For each node, identify credible deviations (for example, more pressure in the column, less cooling to E-104, none flow through P-102, more level in V-101) and document possible causes and consequences.
- **Safeguard review and recommendations**
 - For each deviation, list existing safeguards associated with the T-100 equipment, such as control loops, alarms, interlocks, relief devices, check valves, and physical containment features.
 - Evaluate whether these safeguards are adequate for credible scenarios and propose additional layers of protection where needed, such as high-integrity shutdown systems, improved instrumentation, enhanced operator procedures, or design modifications to T-100, its auxiliaries, or their relief/vent systems.

Overall, the ChatGPT 5.1 AI model did a great job in organizing individual sections to make the sections look presentable. It also used relevant and simple language, which allows the information to be read and understood by people with different levels of experience and background. However, because of the simple language, the AI couldn't depict the sense of danger that comes from this process if it's not conducted properly. In addition, the HAZOP study section isn't what is expected, which usually comes in the form of a table. Additionally, the results obtained from this AI model seem more like a guideline rather than actual results.

Bibliography